

Dear Editors,

We submit the manuscript “Fixed-Budget Local-Patch Risk Routing for Industrial Anomaly Detection” for consideration in Signal, Image and Video Processing (SIViP).

The manuscript studies a practical question in industrial anomaly detection: when the number of local patch computations is fixed, can a risk-aware routing rule allocate that budget more effectively than matched-random selection? Our contribution is deliberately scoped to this question. Across MVTec AD, MPDD, and VisA, the strict matched-random evaluation reports paired category–seed effects with bootstrap confidence intervals, and is compared against a fast-score controller and a nearest-patch-distance dispersion uncertainty controller under an identical fallback count. The main text also reports full-budget detector references, failure cases, exploratory budget sensitivity, online-prefix sensitivity, and latency boundaries. In particular, the manuscript does not claim end-to-end acceleration, universal online robustness, or superiority over full-budget detectors.

We believe the work is appropriate for the journal because it combines a pattern-recognition method with an industrial anomaly-detection evaluation under reproducible, controlled operating conditions. The results are intended to be useful to readers designing fixed-compute inspection systems, especially where the quality of patch selection matters more than an unsupported claim of total runtime reduction.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. A subscription publication route is requested. The source code and reproducibility materials are available in the project repository, and the exact software environment and dataset/license information are documented in the repository and manuscript. There are no undisclosed competing interests.

Thank you for your consideration.

Sincerely,

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on behalf of all authors